

**CENTRE FOR DEVELOPMENT OF IMAGING TECHNOLOGY  
TECHNOLOGY EXTENSION DIVISION**

**SYLLABUS**

| <b>POST GRADUATE DIPLOMA IN COMPUTER APPLICATION<br/>(PGDCA Semester I)</b>                         |                                                     |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Scheme: 2023 Scheme</b><br><b>Duration: 6 Months</b><br><b>Eligibility: A Pass in Any Degree</b> |                                                     |
| <b>THEORY SUBJECTS</b>                                                                              |                                                     |
| <b>PGDCA 101</b>                                                                                    | <b>Introduction to Information System</b>           |
| <b>PGDCA 102</b>                                                                                    | <b>Computer Organisation &amp; Operating System</b> |
| <b>PGDCA 103</b>                                                                                    | <b>Programming in C &amp; Data Structure</b>        |
| <b>PGDCA 104</b>                                                                                    | <b>Relational Database Management System</b>        |
| <b>LAB SUBJECTS</b>                                                                                 |                                                     |
| <b>PGDCA 105</b>                                                                                    | <b>Lab I-C Programming</b>                          |
| <b>PGDCA 106</b>                                                                                    | <b>Lab II- Hardware, Operating System and RDBMS</b> |
| <b>ANNEXURE</b>                                                                                     |                                                     |
| <b>Assignments for Theory &amp; Questions for Lab</b>                                               |                                                     |

## Detailed Syllabus(Theory)

### **PGDCA 101-INTRODUCTION TO INFORMATION SYSTEM**

**Theory Hours - 50 Hrs**

#### **Module I - Computer Fundamentals (10Hrs)**

##### **Learning outcome**

- Understanding of the concept and architecture of computers.
- How a computer system works?
- Learning the basic terminologies.

##### **Topics**

Introduction: - Fundamentals - Definition, Hardware, Software, Operating System, data, information etc; Functions and Characteristics; Applications of Computers; Capabilities and Limitations

Familiarise different generations of computer

Computer classifications: - Analogue / Digital / Hybrid, PC, Microcomputer, Minicomputer, Mainframe computer, Super Computer.

Components of a computer: - Input Unit ; Output Unit ; CPU ; Memory unit

Input Devices:- Keyboard; Mouse; Scanner; Digital/ Web Camera; Touch screen; graphic tablet

Output Devices: – VDU; Printer; Audio Output devices; LCD Projector

Intelligent terminal and dumb terminal

Memory Unit ; Volatile and Non-Volatile memory ; Primary Memory ; Secondary Memory ; BIOS Memory ; Cache Memory ; Virtual memory ; SIMM ; DIMM ; Storage devices; online / offline storage.

Familiarisation of other hardware components: - SMPS; Motherboard; Video Cards - PCI, PCI Express and AGP; Multimedia Cards - Sound Cards, TV Tuner Cards and Capture Cards ; I/O Cards – IDE,SCSI, Serial, USB, and Parallel ; Communications Cards - NICs and Modems

Ports :- Characteristics and different types - Serial, Parallel, USB, Ethernet port etc

Configuring the Peripheral Devices ; methods for connecting peripherals to system - different types of cable connection, explain Bluetooth and Wi-Fi / IR, NFC connectivity

Data representation in computers :- Number System - Types, examples, Conversion rules ; Coding system - ASCII, BCD, EBCDIC, ISCII, Parity Bit, etc.

Units :- Time Units - Milliseconds, Microseconds, Nanoseconds, Pico seconds ; storage units - Bit, Byte, Kilo Byte, Mega Byte, Giga Byte, Tera Byte, Peta Byte, Exa Byte, Nibble

## Module II - Basics of Information Technology (10Hrs)

### Learning outcome

- Acknowledge the role of technologies in today's society
- To identify the basic hardware and software components of a computer and explore their functioning
- Learn about various operating systems

### Topics

**Introduction to IT** - Role of IT in society - Advantages and Disadvantages, Characteristics and features of IT, Application areas - Science and Engineering, Education, Business and Commerce, Medicine and Entertainment - Advantages & Disadvantages of IT

**Basics of Computer System** - Generations & Peripheral devices - Hardware - Software - system software & application software-off the shelf and custom software-programming languages-compilers-interpreters- Operating systems: structure & functions - Types of Operating system: Batch processing OS, Multitasking OS, Multiprocessing OS, Time sharing OS, Real Time OS, Distributed OS, Network OS-Examples of OS: MS-DOS, MS-Windows, Linux Ubuntu, Apple Macintosh

**Introduction to ITES ( IT Enabled Services)** - what are IT enabled services? Benefits and threats, ITES fields - BPO - GIS - Call Centers - Data Management Services - Medical Transcription - Data Digitization, Web services, Payroll management, Telemarketing

## Module III - Basics of Software (10Hrs)

### Learning outcome

- Distinguish System software and Application software.
- Commonly used operating systems.
- Identify the primary functions of Operating Systems.
- Installation and configuration of software.

### Topics

**Types of Software :-** Software - System software , Application software, Utility Software ; Demo ware; Shareware ; Freeware ; Firmware ; Free and Open Source Software.

**Operating Systems :-** Functions, Types - Batch Processing , Single User, Multi User, Multiprogramming, Multi-Tasking (Multiprocessing); Familiarise Multithreading

**Booting and Shutdown of Computers ;** BIOS ; POST ; Master Boot Record(MBR)

Introduction to old versions of Windows OS

**Detailed study of Windows :-** Overview ; GUI basics (Desktop, taskbar, icons, desktop background management) ; Navigation (Start menu, File explorer) ; Accessories ; Control Panel ; User Account management ; storage management ; file and folder management ; keyboard shortcuts ; data backup

**Familiarise Ubuntu Linux :-** Introduction ; History ; Versions ; basics

**Familiarising Command mode :-** Windows (DOS) ; Linux (Terminal)

**Familiarize other operating systems :-** UNIX ; Apple Macintosh

**Linux commands**

**Computer Virus :-** Introduction ; Mechanism of Virus ; How a virus spreads ; How is virus named ; A few prominent viruses ; Types of Computer Virus ; Related Concepts - Antivirus programs - Norton antivirus program

## Module IV : Basics of Information System (10Hrs)

### Learning outcome

- Knowledge about information system, its capabilities and uses with IT infrastructure in a business enterprise to achieve desired goals.

## Topics

**Information System** :- Data, Information & System ; Information System; Dimensions of Information System - Organization, Management, IT ; Functions of Information System

**Business Information Systems** :- Role of Business Information Systems - Top, Middle, Bottom ;

**Types of Business Information Systems** - TPS, MIS, EIS, DSS, AI/Expert Systems

**Management Information System(MIS)** :- Objectives ; Components ; Functions ; Resources ; Characteristics; Terminologies and Applications

**Enterprise System and Types** :- ERP ; CRM ; SCM

## **Module V – Social Media & Cyber Security (10Hrs)**

### Learning outcome

- Understand various social media technologies and their importance in the day to day life.
- Benefits of social media in digital marketing and Education,
- Awareness on various cybercrimes and cyber security,
- Understand copyright and intellectual property rights

## Topics

**Social Media** :- Why & what Social Media – Present and Past ; Social media Platforms and applications ; Usage in business & Education ; social media etiquette

**Social media and digital marketing**

**Information Security** :- Concepts of Security ; Two factor authentication(2FA) ; Biometric authentication methods ; Data Encryption and Decryption ; Cryptography

**Awareness about cybercrime and Cyber security**

**Cyber crime** :- What is cyber crime ; Understand the term computer Virus and Malware ; Email spamming ; Email spoofing ; Email bombing ; Forgery ;Hacking ; Cyber Stalking or Bullying ; Web jacking ; Phishing ; Pornography; Cookies and threats

**Cyber security** :- Basic concepts; Elements of security ; Data security and protection from cyber attack ; Antivirus ; password ; Concept of Firewall ; Digital signature - uses and working

**Cyber Laws :-** An Overview & Scope of Cyber Law ; Copyright ; Trademarks and Software Patenting ; Data Privacy & Confidentiality ; Cyber laws in India ; IT ACT-objectives, features, offences

## **PGDCA 102- COMPUTER ORGANISATION AND OPERATING SYSTEM**

### **Theory Hours - 50 Hrs**

#### **Module I - Processor Organization (10Hrs)**

##### **Learning outcome**

- Familiarize the students with processor organization and Components of Computer.

##### **Topics**

Functional Units of Computer (The Von Neumann Machine)- Computer Function (Fetch and Execute)- Processor organization - Register organization - User visible Registers - Control and Status Registers - Instruction cycle - Data flow - Instruction Pipelining - Pipelining performance- Pipeline Hazards - Parallel Computing - Systolic arrays - Multiple Processor Organization -Symmetric Multiprocessors - Multicore Organization- Stack and Queue - Booting-Motherboard components - Memory slots - Expansion slots - Addon Cards - Chipsets (North bridge and south bridge)- Motherboard form factor - CMOS-SMPS.

#### **Module II - Memory Organization (10Hrs)**

##### **Learning outcome**

- Understand the characteristics of memory systems
- Understand the memory system design and hierarchy
- Understand Different memory accessing technique

##### **Topics**

Define memory - Volatile and Non Volatile memory- Erasable and Nonerasable memory - Distinguish between sequential access and Random Access Memory - Discuss the characteristics of primary memory and secondary memory - Discuss RAM & ROM - Discuss SIMM and DIMM - Memory Hierarchy and its characteristics - Distinguish DRAM and

SRAM - Discuss different types of ROM - PROM, EPROM, EEPROM, EAPROM - Flash memory - Associative Memory - Cache Memory - Cache Performance - Cache Mapping techniques - Types of Cache- Locality of reference - Virtual Memory - Interleaved memory - Discuss the working of Magnetic tape, Hard disk, SSD, Optical storage (Compact disk, DVD, Blue)-Ray disk - RAID levels - History of Cloud Computing - Cloud Architecture - Cloud Storage.

### Module III - Input Output Organization (10Hrs)

#### Learning outcome

- Understand the bus structures in detail and their interconnections.
- Understand Interrupt.

#### Topics

Interrupts and Instruction Cycles - Interrupt cycle - Multiple Interrupts - Describe Bus (Data, Address, Control) and Bus Interconnection - Bus width - Data transfer type - Single Bus Structure - Multiple Bus Structure - Multiple-Bus Hierarchies - Elements of Bus Design - Synchronous and asynchronous bus - Bus Design Parameters - PCI Bus - SCSI Bus - USB Bus - Input Output interface - I/O Bus and interface modules - Asynchronous data transfer - Direct Memory Access - Types of DMA transfer using DMA controller

### Module IV - Operating System (10Hrs)

#### Learning outcome

- Understand various types of Operating Systems.
- Understand various properties of an Operating System.
- Understand various types of process schedulers.
- Understand various scheduling algorithms and Threads

#### Topics

Introduction to Operating System - Functions of Operating System - Type of Operating system, -Batch Operating System, Time sharing Operating System, Distributed Operating System, Network Operating System, Real time Operating system - Operating system services - Operating System Properties, Batch processing, Multi-tasking, Multi

programming - Spooling – Process concept- Process state – Program – Process Control Box  
– Process scheduling queue -Scheduler- Operating System Process scheduling algorithms-  
Thread – Types of threads- Multithreading Models- Introduction to Deadlock - Deadlock  
prevention and avoidance- Kernel I/O Subsystem

## **Module V - Linux Operating System (10Hrs)**

### **Learning outcome**

- Understanding about Linux Operating Systems.
- Understanding about various Linux commands
- Understanding of user management, device management, data management, software installation etc

### **Topics**

Compare Open-source Software and Proprietary Software –Basic Features- Architecture of Linux system- Kernel-Features and Drawback of Linux OS – Linux Distribution -Linux file system- Linux shell types – Essential Linux Command, File handling utilities, Process utilities, Disk utilities Networking Commands- Role of system administrator in Linux – File security and permissions- Installing and removing packages in Linux - Wine on Linux - Understanding of user management, device management, data management, software installation etc in Linux

# PGDCA 103-PROGRAMMING IN C & DATA STRUCTURE

Theory Hours - 50 Hrs

## Module I -Basics of Programming (10Hrs)

### Learning outcome

- Understand the methodology to solve the problem, algorithm, flowchart & coding
- To identify programming language classifications.
- To Learn Procedure and Object oriented programming concepts.

### Topics

Algorithms - Properties of a good algorithm - Flowchart - Problem-solving using Computer -Types of programming (application program, system program, device program etc) Program development Life Cycle - Introduction to Programming Languages - Machine Language - Assembly Language - High Level Programming Languages - Procedure Orientation and Object Orientation - - Compiler and interpreter

## Module II -Introduction to C Programming (10Hrs)

### Learning outcome

- To provide an overview of working principles of C language.
- Construct, compile, link & execute the C programs
- To learn and implement C language programming techniques.

### Topics

Introduction to C programming - Structure of a C program - Input and output Statements -Keywords-Data types - Constants - Variables - Declaring Variable Names - Operators and Expressions - Operator Precedence in Expressions - Increment and Decrement Operators - Conditional Statements - simple if - if-else - switch - Implementing Loops in Programs - TheWhile Loop - The do while Loop - The for Loop - Break and Continue Statement

### Module III - C Arrays and Functions (10Hrs)

#### Learning outcome

- To understand and apply the functions and arrays.
- To implement the features of C language in real world applications.

#### Topics

Arrays - Defining and Manipulating Arrays - Array Variable - Syntax Rules for Arrays - Reading and Writing - Multidimensional Arrays.

Functions - Function Definition - Function Calling - Recursive functions - CharacterStrings - The Character Data Type - Manipulating Strings of Characters - Arrays in Functions.

### Module IV - Pointers and File Handling in C (10Hrs)

#### Learning outcome

1. To understanding pointer concept
2. To Learn File Concept, Reading and Writing operations in Files

#### Topics

Pointer Data Type and its Applications - Pointer Data Type - Pointers and Arrays - Structures and unions - Pointer to structure

Files in C - Creating and Storing Data in a File - Sequential Files - Unformatted Files - Text Files - Pointers and Functions - Preprocessors and Macros creation.

### Module V - Data Structures using C (10Hrs)

#### Learning outcome

- To understand the importance of Data Structure
- Understand the concepts of Searching and Sorting

#### Topics

Data structure introduction - Linear and non Linear data structure - Searching - Linear Search - Binary Search - Sorting - Insertion Sort - Merge Sort - Bubble Sort - Selection Sort - Stack --- Queue - Lists - Linked List - Tree Traversals - Graph

# PGDCA 104 - RELATIONAL DATABASE MANAGEMENT SYSTEM

Theory Hours - 50 Hrs

## Module I - Concepts of Databases (10Hrs)

### Learning outcome

- To understand basic concepts, applications, properties and architecture of database
- Acquire the difference between DBMS & RDBMS
- To learn the fundamentals of data models and to conceptualize and depict a database system using ER diagram

### Topics

**Introduction to Database:-** Database - Introduction, Concept & Overview of Database, Need for database, Advantages, characteristics; Managing Data; File vs Database; Types of Database; Database Users - Database Administrator, Application programmers, Native users; DBMS Architecture; Different levels of databases - internal, conceptual and external levels - data abstraction-database instance and schema; DBMS Components; Database Language - DDL, DML, TCL, DCL; Codd's Rule; ACID Properties in DBMS.

**Database Models:** - Importance; Basic building blocks; Evolution; Types - Relational Data Model, Network Model, Hierarchical Model, Object Oriented Data Models, Semi structured Data model, Entity-Relationship Model

**ER Model-** Basic concepts, Keys, Attributes, ER Diagram, keys, Aggregation, Generalization, Constraints;

**File Organization:** - Indexed, sequential, heap, random, indexed sequential, Hashing; Data Dictionary Storage-metadata-RAID

## Module II - Basics of RDBMS (10Hrs)

### Learning outcome

- To familiarize the concept of relational database structure.
- To demonstrate the use of constraints and relational algebra operations
- To master the importance of normalization in databases

### Topics

**RDBMS:-** Relational Databases - Introduction, Characteristics, Advantages; DBMS vs RDBMS; Data Integrity features - Entity, Referential, User-defined, Domain.

**Relational Database:-** Relational Model concept; Relational Database Structure - Tables, Views, Procedures, Relations, Domains, Attributes, Tuples;

**Relational Algebra:-** Fundamental operations - Select project, Union, Difference, Intersection, Join; Integrity Constraints - Domain, Referential, Entity Integrity, Keys;

**Normalization:-** Functional dependencies; Normal Forms - 1 st , 2 nd and 3 rd Normal forms, BCNF, 4 NF and 5 NF;

**Relational database design principle**

## Module III- Advanced RDBMS Concepts (10Hrs)

### Learning outcome

- To know the fundamental concepts of transaction processing- concurrency control techniques and recovery procedure.

### Topics

**Transaction Processing:-** Transactions – Concepts; Transaction states- active, partially committed, failed, aborted, rollback, transaction isolation

**Concurrency Control:-** Concurrent Transactions; Lock based Protocol; Locks Two-Phase Locking (2PL) ; Time-Stamping Protocol; Optimistic Concurrency Control; Multiple Granularity, Deadlock- detection and Prevention, Recovery from deadlock

**Database Recovery and Security:** - Database Recovery, Kinds of failures, Failure controlling methods; Database errors; Backup & Recovery Techniques; Security & Integrity; Database Security; Authorization;

## Module IV - Structured Query Language (10Hrs)

### Learning outcome

- Learn about SQL – Structured Query Language
- Use the Relational model and how it is supported by SQL
- Perform basic CRUD operations using DML statements, write and call Stored Procedures and Functions stored in the database, write and manage database triggers, cursors, and indexes.

### Topics

**SQL:** - Introduction & Importance; Install a Database Engine; SQL - Syntax, Data Types, Operators, Comments; DDL Commands & Operations; DML Commands & Operations; SQL - Queries, Subqueries, Clauses - DISTINCT, FROM, WHERE, ORDER BY, GROUP BY, HAVING; SQL Aggregate Functions - COUNT, SUM, MAX, MIN, AVG; SQL Conditions - AND, IN, AND or IN, NOT, LIKE, OR, IS NULL, IS NOT NULL, BETWEEN, EXISTS; Set Operations using Unions, Intersect and Except; Constraints - Primary Keys, Unique Constraints, NOT NULL, Foreign Keys; SQL Joins - Inner, Outer, Self; SQL Views & Indexes - Clustered and Non-Clustered Index; DCL Commands & Operations; TCL Commands & Operations; Working with Stored Procedures and Functions, Triggers, Cursors; Backup and Restore

## Module V - Introduction to Database Administration (10Hrs)

### Learning outcome

- Understand, explore and use MySQL Workbench
- Solve Database problems using MySQL
- Gather basic knowledge of advanced database concepts.

### Topic

***Introduction to MySQL*** - Basic introduction; Download and Install MySQL in Windows; MySQL Workbench - Introduction, User Accessibility Options; Workbench Preferences - introduction, SQL preferences, General preferences, Administration preferences, Modeling preferences, others; MySQL Connections - introduction, Client and Server Connections; Introduction to Administrative functions; Workbench Operations; Perform basic CRUD applications using SQL in MySQL

***Advanced Database Concepts:*** - Introduction to Oracle Database Architecture, PL/SQL; Introduction to Hive, Hbase, Hadoop, NoSql, MongoDB, Couchbase

# **PGDCA Semester I Detailed Syllabus (Lab)**

## Detailed Syllabus(Lab)

### **PGDCA 105 -LAB I - C PROGRAMMING**

**Lab Hours - 150 Hrs**

#### **Topics**

##### **Module 1**

1. Exercise programs to understand the procedure of writing algorithms
2. Exercise programs to develop flowchart from the algorithms

##### **Module 2**

1. Exercise programs to understand C tokens and illustrate the working of various Operators.
2. Exercise programs based on Conditional Statements: is, if else, switch.
3. Programming Exercise using loops : for, while, do-while and understand the concept of break and continue statements.

##### **Module 3**

1. Exercise programs to understand Arrays - single dimensional, two dimensional and multidimensional.
2. Exercise programs using functions and recursive functions
3. Exercise programs using arrays in functions
4. Exercise program using string

##### **Module 4**

1. Practice programs on Pointers.
2. Exercise programs using Structures and Unions.
3. Practice programs on File Handling

##### **Module 5**

1. Practice programs to implement various searching techniques in C -linear search, binary search etc
2. Practice programs to create stack and queue
3. Programs to implement tree

## **PGDCA 106 – LAB II- HARDWARE, OPERATING SYSTEM & RDBMS**

### **Lab Hours - 150 Hrs**

#### **Hardware and Operating System**

1. How to configure different peripheral devices.
2. Identify different parts of Mother board
3. Measure the output voltage of SMPS and Check whether SMPS is working or not
4. Identify internal and External Computer ports
5. Installing and Uninstalling different Application Software and antivirus software
6. Setup and install different Computer Printers
7. BIOS Setup Utility and Common Trouble Shooting
8. Understanding control panel settings
9. Different printer servicing and preventive maintenance
10. Identify different Add-on cards used in Personal Computer
11. Identify different parts inside Hard disk and CD drive.
12. Hard Disk Drive formatting and partition.
13. Install and Configure Scanner, Web cam and Biometric device.
14. Identify different RAMs and RAM slots.
15. Assemble and Disassemble the Desktop PC
16. Preventive maintenance of a PC
17. Study of different types of network cable and practically implement the cross-wired cable and straight through cable using Crimping tool
18. Connect the computer in Local area network
19. Operating System Installation (Windows and Linux)
20. Install and Configure Dual Operating System Installation
21. Create Different user account in Windows
22. Temporary disable user accounts
23. Managing user accounts (adding user and deleting user) in Linux
24. Temporary disable user accounts in Linux OS
25. Install, uninstall and update different Application and System software in Linux OS
26. Create Google sheets, Google Docs, Google slides, Google forms and share the link, set the link setting as anyone on the internet with this link can edit

## RDBMS

1. Familiarizations of different database model with example
2. Understanding the installation of database packages
3. Execute programs based on
  - a) DDL commands and operation
  - b) DML commands and operation
  - c) DCL commands and operation
  - d) TCL commands and operation
4. Perform database programs using different constraints
5. Perform database programing using aggregate function
6. Perform different join operations
7. Perform database programs using control structure
8. Perform database programs using procedure
9. Performs database programs using functions
10. Perform database program using trigger
11. Perform installation of MySQL workbench
12. Familiarize with basic workbench operations

# **PGDCA Semester I**

## **Assignments for Theory**

### **and**

## **Questions for Lab**

## **PGDCA I Sem - Paper 1(Theory)**

### **PGDCA 101-INTRODUCTION TO INFORMATION SYSTEM**

#### **Module I - Computer Fundamentals**

##### **Assignment**

1. History and Generations of Computers.
2. Explain in detail about various Input Output devices
3. Explain about different types of storage devices with capacity (based on purpose)
4. Specifications for new personal computer purchase
5. Show the computations needed for the conversion between various number systems.

#### **Module II - Basics of Information Technology**

##### **Assignment**

1. Explain how information systems are related with modern life
2. Role of hardware and software in computing
3. Explain the role of information systems in education
4. Discovering the milestones of ICT (Information and Communication Technology) history

#### **Module III - Basics of Software**

##### **Assignment**

1. Discuss the pros and cons of the three major operating systems.
2. What are the different versions of Windows OS
3. Updating operating system and web browser software
4. Explain about various DOS and Linux commands
5. Various methods for backing up of user data and disc management
6. Computer Virus and Green IT

## **Module IV - Basics of Information System**

### **Assignment**

1. Explain how is an information system is helpful for an organization
2. Explain the role of business information systems
3. Write the advantages of MIS
4. Explain an Enterprise System

## **Module V - Social Media & Cyber Security**

### **Assignment**

1. Explain the elements of cyber security
2. Explain the impact of social media in modern life
3. Explain the working and the advantages of Digital Signature
4. How to market a product using social media?  
How to provide privacy and security features in your different social media accounts

## **PGDCA I Sem - Paper 2 (Theory)**

### **PGDCA 102- COMPUTER ORGANISATION AND OPERATING SYSTEM**

## **Module I - Processor Organization**

### **Assignment**

1. Draw the layout diagram of motherboard
2. Prepare the Configuration of Desktop and Laptop
3. List and explain different types of connectors available in Power supplies
4. Explain about Heatsink and how does a Heatsink work

## **Module II - Memory Organization**

### **Assignment**

1. List and explain different type of hard disk and its features.
2. List different optical disk specification and manufacturer name.
3. Distinguish different types of RAMs.
4. List and explain different shortcut key used in Windows and Linux OS
5. Write a note on Computer virus.
6. Explain the different factors affecting the processing speed of CPU.
7. Explain Symptom of a RAM problem.
8. List Causes of memory damage.
9. Explain Diagnosing memory problems.

## **Module III - Input Output Organization**

### **Assignment**

1. List and use of different external bus used in computer.
2. List and explain different Interrupt used in Computer.
3. Give the note for how does pipelining improve performance of computer architecture.
4. Study of Network Devices Repeater, Hub, Switch, Bridge, Router, Gateway

## **Module IV - Operating System**

### **Assignment**

1. List and explain different Operating Systems
2. Explain the functions of Operating Systems
3. Define the features of Operating System

## **Module V - Linux Operating System**

### **Assignment**

1. Explain the component and architecture of Linux Operating System
2. List and Explain Operating System securities
3. Describe the concepts of virtual memory
4. Make a comparative study between UNIX and LINUX.
5. Make a comparative study between BASH and DOS

## **PGDCA I Sem - Paper 3 (Theory)**

### **PGDCA 103 - PROGRAMMING IN C & DATA STRUCTURE**

## **Module I -Basics of Programming**

### **Assignment**

1. Create Algorithms and flow charts.
  - a) Making instant coffee
  - b) Finding the biggest number among 3 numbers.
2. Programming language classifications and examples.
3. Object orientation examples.
4. Differentiate Compiler and interpreter.

## **Module II - Introduction to C Programming**

### **Assignment**

1. Explain C Tokens.
2. Programs to illustrate the working of various Operators.
3. Explain various condition statements with examples
4. Explain looping statements with example programs.
5. Create a four-function calculator using switch-case.

### **Module III - C Arrays and Functions**

#### **Assignment**

1. Single and multidimensional arrays creation, reading and printing, uses.
2. Explain any 5 string handling functions with example programs.
3. Create a four function calculator with functions add(), sub(), mul(), div()
4. Write a function to find the cube of a number.
5. Write a C program to find the factorial of a number using recursion.

### **Module IV - Pointers and File Handling in C**

#### **Assignment**

1. Explain different File modes and operations in C.
2. What is a pointer? Explain with an example program.
3. Write a short note on Preprocessors and Macros.

### **Module V - Data Structures using C**

#### **Assignment**

1. Explain various searching Techniques with example.
2. Write a short note on Sorting and its Types with examples.
3. Explain Stack and Queue Operations with algorithms for each operation.

## **PGDCA I Sem - Paper 4 (Theory)**

### **PGDCA 104 - RELATIONAL DATABASE MANAGEMENT SYSTEM**

#### **Module I - Concepts of Databases**

##### **Assignment**

1. What is the difference between a logical data model and a physical data model?
2. What are the types of attributes in the ER Model?
3. Discuss Entity-Relationship Diagram in detail.

4. Draw an ER diagram.
5. In a college, a Student enrolls in Courses. A student must be assigned to at least one or more Courses. Each course is taught by a single Professor. To maintain instruction quality, a Professor can deliver only one course
6. Give a network data-structure diagram for the following relational database:  
employee (person name, street, city)  
works (person name, company name, salary)  
company (company name, city)  
manages (person name, manager name)

## **Module II - Basics of RDBMS**

### **Assignment**

1. What are the key ideas behind the relational model? Why are they an improvement over what came before? In what ways is the relational model restrictive?
2. What problems does schema normalization solve? Do you believe that these are important problems?
3. What is the distinction between BCNF and 3NF? Is there a reason to prefer one over the other?

## **Module III - Advanced RDBMS Concepts**

### **Assignment**

1. Define two-phase locking.
2. Explain various types of Locks in Detail.
3. Explain various transaction operations.
4. List the advantages, problems, and applications of the optimistic method of concurrency control.

## **Module IV - Structured Query Language**

### **Assignment**

1. Explain in detail different types of SQL operators?
2. What is the difference between clustered and non-clustered indexes in SQL?
3. What is the difference between the WHERE and HAVING clauses? Explain with an example
4. What is the difference between DELETE and TRUNCATE statements in SQL?
5. Explain character-manipulation functions? Explains its different types in SQL.
6. Consider the below two tables for reference while trying to create queries for the questions asked here.

**Table - EmployeeDetails**

| EmpId | Full Name    | Manager Id | Date Of Joining | City       |
|-------|--------------|------------|-----------------|------------|
| 121   | John Snow    | 321        | 01/31/2014      | Pune       |
| 321   | Kesav        | 986        | 01/30/2015      | Calcutta   |
| 421   | Kuldeep Rana | 876        | 27/11/2016      | New Delhi  |
| 512   | John James   | 456        | 12/02/1999      | Bombay     |
| 625   | Saravanan    | 347        | 23/09/2005      | Madras     |
| 709   | Reghuraj     | 593        | 17/6/2010       | Trivandrum |

**Table - EmployeeSalary**

| Emp Id | Project | Salary | Variable |
|--------|---------|--------|----------|
| 121    | P1      | 8000   | 500      |
| 321    | P2      | 10000  | 1000     |
| 421    | P1      | 12000  | 0        |
| 512    | P3      | 15500  | 2000     |
| 625    | P2      | 7000   | 0        |
| 709    | P3      | 9800   | 1500     |

Write an SQL query

1. To fetch the different projects available from the EmployeeSalary table.
2. To find the employee id whose salary lies in the range of 9000 and 15000.
3. To fetch all the employees who either live in Bombay or work under a manager with ManagerId - 512.
4. To fetch all those employees who work on Projects other than P1.
5. To fetch records that are present in one table but not in another table.
6. To find the maximum, minimum, and average of the employees.

## **Module V - Introduction to Database Administration**

### **Assignment**

1. How are LOOP statements executed in SQL/PL?
2. Explain PL/SQL Block & procedure with examples.
3. List out datatype supported in SQL PL. Explain data type in detail.

4. What will be the output when the rating is 10 AND new\_salary=100? Explain.

```
CASE
WHEN rating >=1 AND rating < 4 THEN
    UPDATE EMPLOYEE
    SET SALARY = SALARY*1.10, BONUS=1000
    WHERE EMPNO=employee_num;
WHEN rating >=4 AND rating<8 THEN
    UPDATE EMPLOYEE
    SET SALARY=SALARY*1.05,BONUS=500
    WHERE EMPNO=employee_num;
ELSE
    UPDATE EMPLOYEE
    SET SALARY=SALARY*1.03,BONUS=0
    WHERE EMPNO=employee_num;
END CASE
```

## **PGDCA I Sem - Paper 5 (LAB)**

## **PGDCA 105 - C PROGRAMMING**

### **Lab Sample Questions**

1. Create an algorithm to compose an email and schedule its sending on a particular date and time.
2. Given the sides of a triangle, write an algorithm to determine its type
3. Develop an algorithm to produce the first 15 numbers of this series:  
1,1,2,3,5,8,13,21...
4. Write Algorithm to swap two numbers without using third variable.
5. Create an algorithm to find the sum of first 100 natural numbers.
6. Create a program to find the given number is even or odd
7. Find the given number is prime or not
8. Create a program to generate electricity bill.
9. Find the reverse of a number
10. Write a program to find the area of a triangle, rectangle or square depending on user's choice.

11. Write a C program to compute the sum of the two given integer values. If the two values are the same, then return triple their sum
12. Write a C program to check whether a given positive number is a multiple of 3 or a multiple of 7
13. Create a program to print Floyd's triangle
14. Write a program to find the roots of a quadratic equation
15. Write a program to convert temperature from Fahrenheit to Celsius
16. Write a program to read values into an array and print them
17. Program to find the transpose of a matrices
18. Program to find the sum of two matrices
19. Program to multiply two matrices
20. Create a function to find the square root of a number
21. Program to find the factorial of a number
22. Program to demonstrate string functions
23. Create functions to find the area of a triangle, rectangle and square using function
24. Write a program in C to read n number of values in an array and display it in reverse order
25. Write a program in C to find the sum of all elements of the array
26. Write a program in C to create and store information in a text file
27. Write a program in C to read an existing file
28. Write a program in C to write multiple lines in a text file
29. Write a program in C to show the basic declaration of pointer
30. Write a program in C to demonstrate how to handle the pointers in the program
31. Write a C program to sort a list of elements using the selection sort algorithm
32. Write a program to implement Binary Search
33. Write a program to create a stack
34. Create a program to implement queue
35. Implement tree using C programming

## **PGDCA I Sem - Paper 6 (LAB)**

### **PGDCA 106 – LAB II- HARDWARE, OPERATING SYSTEM & RDBMS**

#### **Lab Sample Questions**

##### **Session 1- Hardware and Operating System**

1. Perform installation of different peripheral devices like printer, scanner etc using driver software
2. Checking the output voltage from various connectors of SMPS
3. Establish connection between computer and smart TV
4. Download and install any free software available on internet
5. In BIOS setup pen drive as boot device
6. Replace 2 GB RAM with 4GB RAM and understands the changes in computer speed
7. Partition and format HDD to C,D drives and make some spaces free
8. Install Windows OS in C drive, and install Linux in free space
9. Create user account and set password in Windows and Linux OS
10. Perform computer assembling
11. Perform trouble shooting
12. Establish a wired internet connection in your computer
13. Install a peripheral device via control panel

##### **Session 2-RDBMS**

###### **Exercise 1**

Consider the following schema for a Library Management Database:

BOOK (Book\_id, Title, Publisher\_id, Publication\_Year, ISBN\_Code, Category\_id,  
Department\_id, No.of Copies, Edition, Price, Status)

BOOK\_AUTHORS (Book\_id, Author\_Name)

PUBLISHER (Publisher\_id, Name, Address, Phone)

Department (Department\_id, Department\_Name)

Category(Category\_id, Category\_Name)

Staff\_Details (Staff\_Id , Staff\_Name, Post, Department\_id, contact\_Number, Card\_no)

Library\_Staff(LibStaff\_Id, LibStaff\_Name, Password, Is\_Admin, Designation)

Student\_Details (Student\_Id , Student\_Name, Sex, Date\_Of\_Birth, Course, Department\_id, contact\_Number, Card\_no)

Book\_Issue(Card\_no, Book\_id, Issue\_Date, Due\_Date, LibStaff\_Id, Fine)

1. Draw E-R diagram and convert entities and relationships to relation table for a given scenario.
2. Develop a schema for a given scenario.
3. Perform Normalization till 4th Normal Form (covering 1NF, 2NF, 3NF, BCNF, and 4NF) on the database system for library management system
4. For the Library Management Database, Write SQL queries to
  - a) Retrieve details of all books in the library – id, title, name of publisher, authors, number of copies in each category, etc.
  - b) Get the particulars of borrowers who have borrowed more than 3 books, but from June 2020 to Mar 2021
  - c) Delete a book in BOOK table. Update the contents of other tables to reflect this data manipulation operation.
  - d) Partition the BOOK table based on year of publication. Demonstrate its working with a simple query.
  - e) Create a view of all books and its number of copies that are currently available in the Library.
5. Learn to perform simple db operations using Hive, Hbase, Hadoop, NoSql, MongoDB, Couch base

## Exercise 2

Consider an orders table with the following structure.

| Order_id | Order_date | Order_amt | Quantity | Cust_name | Salesman_id |
|----------|------------|-----------|----------|-----------|-------------|
| 1        | 12/22/2021 | 160       | 2        | Smith     | 2           |
| 2        | 08/10/2021 | 190       | 2        | Johnson   | 5           |
| 3        | 07/13/2021 | 500       | 5        | Bonny     | 1           |
| 4        | 07/15/2021 | 420       | 2        | Smith     | 2           |
| 5        | 12/22/2021 | 1000      | 4        | Winsam    | 3           |
| 6        | 10/02/2021 | 820       | 4        | Smith     | 5           |
| 7        | 11/03/2021 | 2000      | 2        | Bonny     | 4           |
| 8        | 09/09/2021 | 1560      | 3        | Ruby      | 1           |
| 9        | 03/06/2021 | 50        | 10       | Roy       | 2           |
| 10       | 04/04/2021 | 60        | 5        | Ruby      | 2           |
| 11       | 03/13/2021 | 80        | 8        | Smith     | 3           |
| 12       | 08/13/2021 | 4500      | 3        | Bonny     | 2           |
| 13       | 06/27/2021 | 3500      | 1        | Roy       | 5           |
| 14       | 11/03/2021 | 450       | 6        | Winsam    | 3           |
| 15       | 03/06/2021 | 20        | 50       | Ronson    | 1           |

Now, based on the above concepts write queries for the following

1. Count the number of orders made by a customer with Customer name Smith made.
2. Write a SQL query to calculate average order amt of all orders. Return average amount.
3. Write a SQL query to count the number of unique salespeople.
4. Write a SQL query to count the number of customers. Return number of customers.

5. Write a SQL query to find the maximum order amount.
6. Write a SQL query to find the minimum order amount.
7. Write a SQL query to find the highest purchase amount ordered by each customer.  
Return customer\_name, maximum order amount.
8. Write a SQL query to find the highest purchase amount ordered by each customer on a particular date. Return, order date and highest order amount.
9. Write a SQL query to find the highest purchase amount on '03/06/2021' by each sales person. Return salesperson ID, order amount.
10. Write a SQL query to find highest order amount by each customer in a particular order date. Filter the result by highest order amount above 1500.00. Return customer id, order date and maximum order amount.
11. Write a SQL query to find the maximum order amount generated by each sales person. Filter the rows for the salesman\_id is in the range 2 and 4 (Begin and end values are included.). Return sales person id and maximum purchase amount.
12. Write a SQL query to count number of products where order price is higher than or equal to 350. Return number of products

### **Exercise 3**

From the following table, write a SQL query to compute the average price for unique companies. Return average price and company id.

| PRO_ID | PRO_NAME         | PRO_PRICE | COM_ID |
|--------|------------------|-----------|--------|
| 101    | Mother Board     | 3200.00   | 15     |
| 102    | Key Board        | 450.00    | 16     |
| 103    | ZIP drive        | 250.00    | 14     |
| 104    | Speaker          | 550.00    | 16     |
| 105    | Monitor          | 5000.00   | 11     |
| 106    | DVD drive        | 900.00    | 12     |
| 107    | CD drive         | 800.00    | 12     |
| 108    | Printer          | 2600.00   | 13     |
| 109    | Refill cartridge | 350.00    | 13     |
| 110    | Mouse            | 250.00    | 12     |

#### **Exercise 4**

From the following tables write a SQL queries

Table : emp\_department

| DPT_CODE | DPT_NAME | DPT_ALLOTMENT |
|----------|----------|---------------|
| 57       | IT       | 65000         |
| 63       | Finance  | 15000         |
| 47       | HR       | 240000        |
| 27       | RD       | 55000         |
| 89       | QC       | 75000         |

Table : emp\_details

| EMP_IDNO | EMP_FNAME | EMP_LNAME | EMP_DEPT |
|----------|-----------|-----------|----------|
| 127323   | Michale   | Robbin    | 57       |
| 526689   | Carlos    | Snares    | 63       |
| 843795   | Enric     | Dosio     | 57       |
| 328717   | Jhon      | Snares    | 63       |
| 444527   | Joseph    | Dosni     | 47       |
| 659831   | Zanifer   | Emily     | 47       |
| 847674   | Kuleswar  | Sitaraman | 57       |
| 748681   | Henrey    | Gabriel   | 47       |
| 555935   | Alex      | Manuel    | 57       |
| 539569   | George    | Mardy     | 27       |
| 733843   | Mario     | Saule     | 63       |
| 631548   | Alan      | Snappy    | 27       |
| 839139   | Maria     | Foster    | 57       |

1. To display all the data of employees including their department.
2. To display the first name and last name of each employee, along with the name and sanction amount for their department
3. To find the departments with a budget more than Rs.50000 and display the first name and last name of employees.
4. To find the names of departments where more than two employees are working.  
Return dpt\_name